

Source of funds, *by construction*

MPS-0025 asks how a custodian can obtain provenance evidence for an incoming shielded asset. Custom spend logic answers upstream of that: the chain proves compliance since mint, and the custodian's job becomes choosing which token types are guarded by contracts its lawyers accept.

READS AGAINST

[MPS-0025](#) — Source-of-Funds Evidence for Incoming Shielded Assets

SOURCE DESIGN

Custom spend logic — early MIP draft, direction only

ALSO TOUCHES

[MPS-0013](#), [MPS-0024](#), [MPS-0021](#), [MPS-0014](#), [MPS-0029](#)

STATUS

Draft assessment,
September 2026

01 The gap MPS-0025 names

[MPS-0025](#) separates three things that a transparent chain lets a custodian conflate.

Receipt — the asset arrived, and the ledger will confirm it. **Attribution** — where the asset came from, which the shielded ledger deliberately does not say. **Acceptance** — whether the asset can be credited, redeemed, passed on, quarantined or rejected, which is the decision the custodian actually has to make and cannot make from receipt alone.

Every mechanism proposed against that gap so far has tried to supply the missing middle term. Reconstruct attribution — through an attestation, an encrypted transfer context, a selective disclosure, a viewing capability — and acceptance follows from it. That framing is what MPS-0025's seven open questions are all shaped around: who attests, in what envelope, verified how, bound to which asset.

Custom spend logic makes acceptance decidable without attribution — and then, where a custodian does need attribution, lets the rule demand it as a condition of the spend. Either way the decision moves off the arriving asset and onto the asset class, taken once, in advance, by the people qualified to take it.

02 Two kinds of evidence

The two families have different failure modes, and [MPS-0025](#)'s requirements land very differently on each.

FAMILY A

Evidence *about* a transfer

Something is produced alongside the asset and travels with it: an attestation, an encrypted envelope, a disclosed viewing capability, a bilateral message between institutions.

- Depends on the sender producing it honestly and completely.
- Needs a binding argument — that *this* evidence describes *this* coin.
- Says nothing about hops before the one it describes.
- Nothing prevents the non-compliant transfer; it is visible afterwards, if at all.

Answers "what do I know about where this came from?"

FAMILY B — THIS DESIGN

Evidence *by* construction

A rule is bound into the token type at mint. The Zswap spend circuit re-derives the type from the rule on every spend and refuses to produce a valid input unless the rule holds. A coin of that type therefore cannot exist unless the rule held on every hop that led to it.

- Independent of sender cooperation, honesty, or availability.
- No binding problem — the evidence *is* the coin.
- Covers the entire history, not one hop.
- The non-compliant transfer never happens; there is nothing to detect.

Answers "could this have reached me any other way?"

The apparent trade is that Family B says nothing specific. It does not name a sender, an institution, or a customer; it certifies that a stated rule was enforced, and the rule is whatever the issuer wrote at mint. That reads as a hard limit — a custodian needing an actual fact about an actual counterparty would seem to be back in Family A.

A rule can *compel* Family A evidence, and compelled evidence is worth more than volunteered evidence.

03 Where the guarantee comes from

The whole property rests on one move: the spend rule is an operand of the token type's preimage, so it cannot be detached from the asset without producing a different asset.

AT MINT

The rule becomes part of the type

contract address

domain separator

spend rule

Today a token type is `SHA256("midnight:derive_token" || domain_sep || contract address)`: the mint effect publishes the raw, contract-chosen `domain_sep` and the ledger derives the type at validation. The design extends that preimage with a third operand, the spend rule descriptor, encoded empty for an ungarded type so existing types derive to the same value. Address and separator are already public, so an indexer can build `type → (contract, separator)` today; the `type → rule` mapping awaits the published descriptor, a real ledger change.



AT EVERY SPEND

The circuit re-derives and enforces

witness descriptor

re-derive type

enforce rule

The Zswap `spend` circuit takes the descriptor as a private witness, recomputes the type, asserts it matches the coin, then enforces the rule. That assert runs on every shielded spend, so a guarded coin is reachable only through its rule. Where the descriptor commits to the rule's own logic — a mode-2 verifying key — substituting the rule substitutes the asset. A mode-3 descriptor names an address, and pins only that it was consulted.



AT RECEIPT

The coin's existence is the evidence

coin of type T

A custodian holding a coin of type `T` checks `T` against the types it has approved. The guarantee runs unbroken back to the mint — there is no hop at which the rule could have lapsed, because a coin that skipped it would not be a coin of this type. Under mode 3 what runs unbroken is the consultation, not a fixed verdict.

Everything expensive happens on the spend path, and it happens to the sender. What is left for the receiver is a hash comparison.

04 Three modes, three postures toward a custodian

From [MPS-0025](#)'s side the design's three authorisation modes are three different bargains between how much a rule can check and how much the transfer discloses.

MODE 1

TODAY

No rule

The current behaviour, with nothing added: the spend is authorised exactly as it is today. Every token type in existence resolves here.

WHAT A CUSTODIAN GETS

Nothing beyond today. Receipt without attribution — this is the state MPS-0025 describes.

COST

None. It is the baseline.

MODE 2

NEEDS [MPS-0014](#)

Recursive verification of a pure circuit

The type names a gate circuit's verifying key; the `spend` proof recursively verifies a proof against it. Nothing becomes public.

WHAT A CUSTODIAN GETS

Enforced statements the spender can prove from what they already hold — an attested credential, a verified counterparty, a jurisdiction claim — and a disclosure envelope the spender must produce and bind before the spend is valid.

COST

Blind to chain state. Cannot consult a freeze list, a denylist, or a sanctions set until the ledger exposes a provable state root.

MODE 3

GENERAL FORM

A contract call bound to the spend

The type names a contract address and entry point; the spend is well-formed only if the same transaction carries a call to it — the authorising call lives in an intent — bound by a new field analogous to the communication commitment, which the `spend` proof opens: the caller/callee binding contract-to-contract calls already use, with the spend in the caller position.

WHAT A CUSTODIAN GETS

Everything an AML programme actually asks for — live list checks, threshold authorisation,

public auditable events, rules the issuer can upgrade behind a governed pointer without reissuing the token, an upgradeability that cuts both ways — and an *attested* spender. The `spend` proof already witnesses the spending secret key, so it can prove the `ZswapCoinPublicKey` it asserts is the one behind this nullifier; the gate reads a proved caller rather than a forgeable `ownPublicKey()` witness.

COST

The contract call is public, so the asset's token type is visible on every spend, and free merging of offers is likely lost with it. The same pointer that makes a rule upgradeable means the type pins which contract is consulted, not what it will decide: the no-substitution guarantee is mode 2's. The binding needs no new cryptography — it reuses the primitive contract-to-contract calls run on today — but the fields and checks it needs do not exist yet.

For MPS-0025's representative use cases the split is clean. A rule that must consult an issuer's list — the sanctions and eligibility checks behind Use Case 1 and Use Case 5 — needs mode 3, because only a contract can read live state the spender must not see. A rule that only has to bind the sender's own credentials to the spend — Use Case 2's institution-to-institution attribution — fits mode 2, though the receiving institution still needs a channel for the disclosed identity itself; the rule can make the envelope mandatory but does not transport it. Each needs the spending wallet to obtain and run a gate it has never held, for whatever type it happens to hold. Mode 3's gate is a contract whose local state the spender must not see, so it rests on the capsule runtime; mode 2's gate is a bare circuit, so what it needs instead is a chain-resolvable source for the gate's proving artefacts — plausibly the ZKIR slot beside an entry point's verifier keys, itself unsettled.

05 Attribution, when the rule asks for it

The two families compose. A spend rule is a predicate over the spend, and one of the things a predicate can demand is that the spender *produce* something — so a rule can

require a ciphertext to be created and bound to the spend before the spend is valid at all.

That does not replace Family A evidence; it changes its character. Today an attestation or an encrypted transfer context is voluntary. The sender produces it because a bilateral agreement, an internal policy, or goodwill says so, and a recipient who does not get one has no recourse beyond rejecting the asset after it has already arrived. Under a rule, producing it is a condition of the coin moving at all. A sender who omits the envelope has not made a non-compliant transfer — they have failed to transfer.

GRADED DECISIONS

Quarantine and escalate need something to score, which a binary rule-held-or-not guarantee cannot give them. A mandated envelope can carry exactly the fields a custodian's risk model reads.

AUDIT RETENTION

The envelope is the per-asset record a custodian has to keep, and under a rule it exists for every transfer rather than only the ones a counterparty chose to document.

NAMED COUNTERPARTIES

Institution-to-institution attribution needs the sender's identity *delivered*, not merely enforced, and a rule can compel that delivery. What a mode-3 gate adds is the harder half: a sender identity *proved* against the nullifier rather than self-asserted.

Where the ciphertext is published

One candidate already exists in the protocol: each Zswap circuit accepts an arbitrary bound input, whose exact value the proof verifies against. That is how the `output` circuit binds its ciphertext today, and the slot has no use yet in the `spend` circuit. Whether that, a mode-3 contract's public state, or an off-chain carrier with only a commitment on the spend path is the right answer is open.

Who can open it

The recipient, the issuer, a designated authority, or several parties in one envelope — an issuer's choice, expressed in the rule, with a materially different privacy consequence each way. Key rotation over the life of a token type is its own unanswered question.

What it costs

Anything published per spend is a per-spend artefact that an observer can count, size, and correlate even when they cannot read it. Mandatory disclosure buys attribution with metadata, and the price is paid on every transfer of the asset, not only the ones anyone examines.

The point for [MPS-0025](#) is the ordering. Attribution stops being the mechanism and becomes a payload the mechanism can require — which means the questions MPS-0025 asks about envelopes, attestors and binding do not go away, but they stop being the load-bearing part of the design and become parameters an issuer sets per token type.

06 Against MPS-0025's core requirements

[MPS-0025](#) lists eight properties a source-of-funds model should preserve. Assessed against the design as drafted, with its open questions unresolved.

R1

Sufficient provenance evidence for an incoming asset

Met, in a different currency than MPS-0025 assumes. The custodian gets a guarantee that a specific rule was enforced on every hop since the mint, not a description of the origin. Sufficiency therefore stops being a per-asset judgement and becomes a legal reading of the guarding contract: it holds exactly when that contract encodes what the custodian's own obligations require — and for a mode-3 contract that reading has to cover who may change it, since the type fixes the address and not the logic behind it.

R2 **Supports accept / reject / quarantine / escalate**

Partial. Accept and reject fall straight out of the binary guarantee — the rule held or the coin does not exist. Quarantine and escalate need something to grade, which a rule can supply by mandating a disclosure envelope; the mechanism is there, but what the envelope carries and where it is published are undecided, so the graded half of R2 is a direction rather than a result.

●●○ PARTIAL

R3 **Does not make all shielded transfers publicly traceable**

Met. Nothing becomes traceable. Modes 1 and 2 add no public data at all; mode 3 publicly attaches a gate call to a spend, which reveals the token type of that spend but never the recipient, amount, or transfer graph. Rules are opted into per token type by the issuer.

●●● MET

R4 **Sender, recipient, amount, graph remain shielded by default**

Met under modes 1 and 2; degraded under mode 3, where the visible contract call narrows the anonymity set for that asset to holders of token types sharing the rule, and where whether the sender identity the gate is handed is published or opened only to the gate is undecided. This is the design's sharpest privacy cost and it is a per-issuer choice, not a protocol default.

●●○ MET, MODE-
DEPENDENT

R5 **Disclosure limited to the parties that require it**

Met, and arguably the strongest fit. Nothing is disclosed to anyone. There is no envelope to scope, no key to hand over, no recipient list to get wrong, and no way for the evidence to be over-shared because the evidence never leaves the coin.

●●● MET

R6 **Custodian can retain audit evidence**

Partial. What is retainable and independently reproducible is strong: the token type, the public mint effect that fixed its preimage, and the rule's own source. The per-asset record of why *this* asset was accepted can come from a rule-mandated envelope — and unlike a voluntary one it would exist for every transfer — but the

carrier is unspecified. Mode 3's public event is available to anchor it in the meantime.

●●○ PARTIAL

R7 **No bespoke manual agreement per inbound transfer**

Met, and this is the operational payoff. Diligence moves from per-transfer to per-token-type: a custodian reviews an asset's rule once, at onboarding, and every subsequent inbound coin of that type is decided by a type comparison. No counterparty coordination, no bilateral evidence protocol. Once is literal only where the rule is immutable; a governed mode-3 contract has to be watched, not merely read.

●●● MET

R8 **Permissionless shielded transfer stays usable**

Met by construction. Mode 1 is today's behaviour, and every existing token type is a mode-1 type that recomputes unchanged under the extended derivation. A token acquires a rule only because its issuer chose one at mint; nothing is imposed on the unregulated ecosystem.

●●● MET

●●● the design delivers this as specified ●●○ delivered with a stated condition or cost
○○○ outside what this mechanism reaches

07 **Against MPS-0025's open questions**

Several of the seven are answered. Several are dissolved rather than answered — the design makes the question stop applying, which is a different thing from solving it.

Q1 **What mechanism yields evidence without making transfers traceable?** —

ANSWERED

Bind the rule into the token type and enforce it on the spend path. Nothing about the transfer becomes traceable because nothing about the transfer is published — the evidence is structural, carried by the fact that a coin of that type exists at all.

Q2 **Embedded, encrypted, out-of-band, selectively disclosed, or attested?** +

Q3	How does the custodian verify the evidence corresponds to this specific asset?	+
Q4	Which party produces or attests to the evidence?	+
Q5	What protocol, SDK, wallet, indexer and custody support is required?	+
Q6	How to support regulated inbound flows without breaking permissionless ones?	+
Q7	What is the minimum viable flow a compliance team could approve?	+

08 Where it does not reach

Five limits are structural, and each is a reason [MPS-0025](#) does not close on this design alone.

It cannot be retrofitted onto an issued asset

Because the rule is part of the type, an asset already in circulation cannot acquire controls. The issuer must mint a new type and migrate holders — and migration needs the very controls the old type lacks, since nobody can compel a holder to move. Any asset that may ever face a source-of-funds requirement has to be issued with a rule from the outset, even a permissive one a governed contract can later tighten.

The disclosure carrier is named as a need, not designed

A rule can make attribution mandatory, which is the hard half. What the draft does not settle is where the ciphertext lives, who holds the keys, how they rotate over a token type's life, and what the per-spend artefact costs in metadata. Until those are answered, every MPS-0025 requirement that leans on a delivered payload — the graded decisions, the retained record, named counterparties — is supported in principle and unbuilt in practice.

It has nothing to say about uncontrolled assets

Use Case 3 — an asset arriving with insufficient evidence — is not solved, it is bifurcated. For a controlled token type the situation cannot occur. For everything else the custodian is exactly

where MPS-0025 found it, and the quarantine workflow is still needed. A custodian's inbound policy therefore becomes two policies, not one.

The strongest mode costs the asset's type privacy

Mode 3 is where the AML-grade rules live, and it attaches a publicly visible gate call to every spend. Holders of a mode-3 asset should understand that they are trading that asset's transaction-graph privacy for its acceptability, permanently and for every transfer. Closing that gap is what the merklized state root sketched late in the specification is for, and it is not yet designed.

Mode 3's binding is sketched, not specified

A gate that only observes the nullifier through effects establishes co-presence and nothing about who spent — over a wallet-held coin it cannot derive that nullifier itself — so the binding must originate in the Zswap `spend` circuit. That binding needs no new cryptography, but it needs parts that do not exist: a proved field on the shielded `Input` and a position for it in the spend proof, a public-address shape for a shielded caller, a shielded branch in the `caller` derivation — a shielded-only intent yields no caller at all today, which [MPS-0029](#) already lists as open — and a ledger check strong enough to make a spend malformed *without* its authorisation, where the existing claimed-call test is only a subset. The routing is unresolved too: binding a call to a spend means naming publicly which rule to run, while the shielded input's type is deliberately never disclosed — mode 2, fully in-circuit, is unaffected. New public inputs also move the `spend` circuit's verifier keys, which is hard-fork-scale.

09 What changes if this is the answer

Adopting this as [MPS-0025](#)'s direction moves work, rather than adding it. Three consequences are worth putting in front of whoever decides.

DILIGENCE SHIFTS

From per transfer to per token type

The object of review becomes the contract guarding a token type — a public, readable artefact that a legal and compliance team can assess the way it assesses any other control, before any exposure. It is fixed if the type commits to a mode-2 circuit; a mode-3

TIMING SHIFTS

From acceptance time to issuance time

Whether an asset is acceptable to regulated holders becomes a property fixed when it is minted. That is a real constraint on issuers: the decision is irreversible and its consequences land on holders years later. What mint fixes under mode 3 is the address consulted, not

address makes that review standing rather than one-off. What the assessment produces is an allowlist of token types. Inbound coins are then decided by a hash comparison, with no evidence protocol running on arrival at all.

the rule it runs — irreversible and mutable at once.

DEPENDENCY SHIFTS

Onto resolving gate code

A spender's wallet has to obtain and run a gate it has never held, for an arbitrary token type; without a chain-resolvable source that means out-of-band delivery of gate code and keys. The rules custodians actually need read state the spender must not see, so a mode-3 gate is a contract, and the Capsule Runtime MIP — which answers [MPS-0021](#) with encapsulated contract execution and local state — resolves it through its executable registry. A mode-2 gate is a bare circuit, and what it needs is a chain-resolvable source for its proving artefacts. Neither supplies the attested sender: a capsule's secret is a per-contract witness, not a coin key. That gap is [MPS-0029](#)'s.

Reads an early, direction-only design draft against a proposed MPS

Only the draft's Specification is worked; everything else in it is still template

Every verdict here will move as either document changes